

PNEUMOCOCCUS

Streptococcus pneumoniae

Microbiology (Bacteriology) • Ananthanarayan & Paniker, Chapter 23

1. INTRODUCTION

- *S. pneumoniae* is a **Gram-positive, lanceolate-shaped diplococcus**, a normal inhabitant of the upper respiratory tract.
- Differs from other streptococci by its **morphology, bile solubility, optochin sensitivity** and a **specific polysaccharide capsule**.
- Single most common bacterial cause of **pneumonia** and **otitis media** in children; also sinusitis, bronchitis, bacteraemia, meningitis.
- First seen by **Pasteur and Sternberg (1881)**; association with pneumonia shown by **Fraenkel and Weichselbaum (1886)**.

2. MORPHOLOGY

- Small (~1 μm), slightly elongated cocci, one end broad, other pointed — **flame-shaped / lanceolate**.
- In **pairs (diplococci)** with the **broad ends in apposition**; long axis parallel to the line joining the pair.
- **Capsulated** — capsule encloses each pair; best seen in exudates, lost on repeated culture. In culture: more rounded, may form short chains.
- Non-motile, non-sporing, **Gram-positive**. Capsule shown as a clear halo in **India ink** preparation.

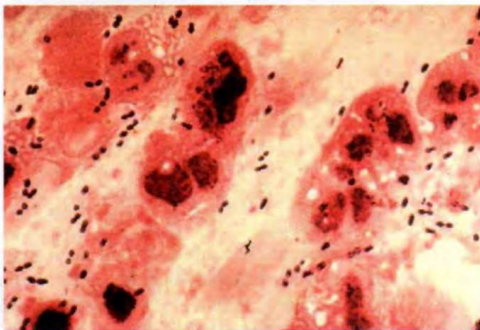


Fig. 23.1 Gram-stained smear of sputum containing Gram-positive diplococci

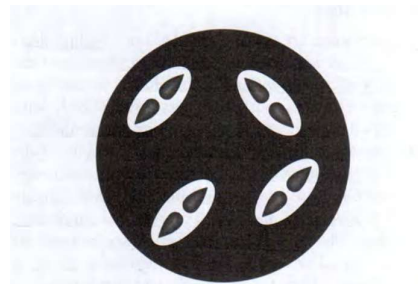


Fig. 23.2 *S. pneumoniae*; India ink preparation to show capsules

3. CULTURAL CHARACTERISTICS

- Complex growth requirements — grow only in **enriched media**. Aerobes and facultative anaerobes; optimum **37 °C** (25–42 °C), **pH 7.8**; growth improved by **5–10% CO₂**.
- **Blood agar (18 h)**: small (0.5–1 mm), dome-shaped, glistening colonies with **green discolouration (α -haemolysis)** — like viridans streptococci.
- On further incubation colonies become flat with raised edges and **central umbonation** → concentric rings = "**draughtsman**" / "**carrom coin**" colonies (due to autolysis).
- Types **3 and 7** (abundant capsule) → large **mucoïd** colonies. Under **anaerobic** conditions → **β -haemolysis** (oxygen-labile haemolysin O).
- **Liquid media: uniform turbidity**. Cultures readily undergo **autolysis** (intracellular enzymes) — enhanced by **bile salts**, sodium lauryl sulphate; heat-killed cultures do not autolyse.

4. BIOCHEMICAL REACTIONS & IDENTIFICATION TESTS

- **Catalase and oxidase negative**. Sugar fermentation in **Hiss's serum sugars** — **inulin fermented** (streptococci do not).

Bile solubility test

- Few drops of **10% sodium deoxycholate** added to 1 ml overnight broth culture → culture **clears** (lysis); or a loopful on a colony lyses it in minutes. Test at **neutral pH** with live young cells.

- **Principle:** bile activates an **autolytic amidase** that cleaves the bond between **alanine and muramic acid** in peptidoglycan. Bile solubility is a **constant property** — of great diagnostic value.

Optochin sensitivity

- **Optochin (ethyl hydrocupreine) 1/500,000** disc on inoculated blood agar → **wide zone of inhibition** with *S. pneumoniae*; viridans streptococci are resistant.

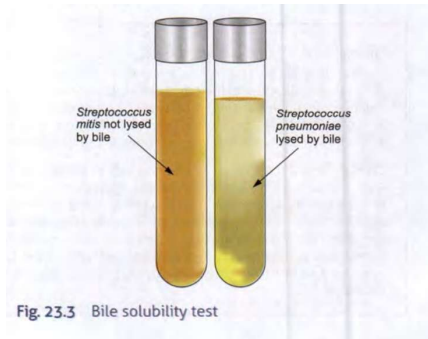


Fig. 23.3 Bile solubility test



Fig. 23.4 *S. pneumoniae* colonies sensitive to optochin

Feature	<i>S. pneumoniae</i>	Viridans streptococci
Morphology	Capsulated, lanceolate diplococci	Non-capsulated, oval/round cells in chains
Quellung test	Positive	Negative
Colonies on blood agar	Dome-shaped → draughtsman	Dome-shaped
Liquid media	Uniform turbidity	Granular turbidity, powdery deposit
Bile solubility	Positive	Negative
Inulin fermentation	Positive	Negative
Optochin sensitivity	Sensitive	Resistant
Intraperitoneal inoculation in mice	Fatal infection	Non-pathogenic

Table 23.1 of the textbook.

5. RESISTANCE

- Delicate: **thermal death point 52 °C for 15 min**; die on prolonged incubation (toxic peroxides). Maintained on semi-solid blood agar or by **lyophilisation**.
- β -lactams are drugs of choice; all strains sensitive to 0.05 mg penicillin until **1967**, when resistant strains appeared.

6. ANTIGENIC PROPERTIES & VARIATION

- **Capsular polysaccharide** — most important, **type-specific** antigen; diffuses into medium/exudates → **specific soluble substance (SSS)**. >90 serotypes (types 1, 2, 3...).
- **Serotyping:** agglutination with type-specific antiserum; precipitation of SSS; **capsule swelling / Quellung reaction (Neufeld, 1902)** — suspension + type-specific antiserum + methylene blue → capsule appears **swollen, sharply delineated and refractile**. Can be done directly on sputum.

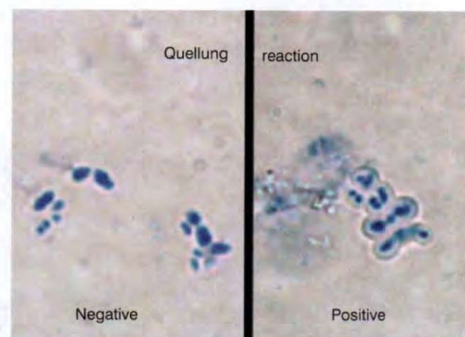


Fig. 23.5 Quellung reaction

- **Other antigens:** species-specific **nucleoprotein** and **somatic "C" carbohydrate**.

- **C-reactive protein (CRP):** abnormal **beta globulin** in acute-phase sera that precipitates with the C antigen; **not an antibody** — an **acute-phase protein made by hepatocytes**, raised in infection, inflammation, malignancy, tissue destruction; disappears as inflammation subsides.
- **S-R variation:** on repeated subculture, rough forms — **non-capsulated, autoagglutinable, avirulent**.
- **Transformation (Griffith, 1928):** rough pneumococci acquire capsules of another serotype when treated with its DNA — **first demonstration of genetic exchange in bacteria**.

7. VIRULENCE FACTORS

- **Capsular polysaccharide** — acidic, hydrophilic; **inhibits phagocytosis** (though "surface phagocytosis" against fibrin/epithelium still occurs). Type 3 most capsule → high virulence; non-capsulated strains **avirulent**; anticapsular antibody is **protective**.
- **Pneumolysin** — membrane-damaging toxin, cytotoxic and **complement-activating**; immunogenic. **Autolysins** — release bacterial components in tissues.
- Oxygen-labile haemolysin and leucocidin — weak, little role.

8. PATHOGENICITY & DISEASES

- Experimentally: intraperitoneal injection kills **mice/rabbits in 1–3 days**; organisms in peritoneal exudate and heart blood.
- Colonise nasopharynx → spread to **middle ear, sinuses, respiratory tract**; meninges by contiguity or blood; bacteraemia → heart, peritoneum, joints. Usually **endogenous**.
- **Lobar pneumonia:** aspiration of nasopharyngeal secretions when defences are impaired (viral infection, anaesthesia, chilling) → spread along peribronchial tissues and lymphatics; early **bacteraemia**; toxemia due to diffusion of capsular polysaccharide. **Crisis** (fall in temperature) coincides with neutralisation of SSS by anticapsular antibody. **Type 3** most virulent.
- **Bronchopneumonia:** almost always **secondary**; any serotype; often terminal in aged/debilitated. **Acute exacerbations of chronic bronchitis** (with *H. influenzae*).
- **Meningitis** — **most serious**; usually secondary to pneumonia, otitis, sinusitis or conjunctivitis; all ages; untreated nearly always fatal; case fatality ~25% even with treatment.
- Commonest infections: **otitis media & sinusitis**. Others: empyema, pericarditis, suppurative arthritis, peritonitis, keratitis, dacryocystitis.

9. EPIDEMIOLOGY

- Source: respiratory tract of **carriers** (about **half the population** at any time) and patients; spread by **droplets / droplet nuclei**; crowding helps.
- Disease when resistance is lowered: viral respiratory infection, pulmonary congestion, malnutrition, immunodeficiency, **alcoholism, splenectomy, sickle cell disease**.
- Adults: types **1–8** cause ~75% of pneumonia; children: types **6, 14, 19, 23**. Commoner in **winter** and at the two extremes of age; epidemics in closed communities (army camps).

10. LABORATORY DIAGNOSIS

- **Specimen:** sputum, CSF, blood (culture); urine (antigen detection).
- **Microscopy:** **rusty sputum** in lobar pneumonia — Gram-positive lanceolate diplococci in large numbers, few other bacteria. **CSF** — diplococci **inside polymorphs** and extracellularly.
- **Culture:** blood agar, 37 °C, **5–10% CO₂**; **gentamicin (5 µg/ml) blood agar** for respiratory samples. **Blood culture** in glucose broth — isolation from blood indicates **bad prognosis**.
- **Identification:** α-haemolysis, draughtsman colonies, catalase negative, **optochin sensitive, bile soluble**, inulin fermented, **Quellung +**.
- **Mouse inoculation:** for scanty organisms — mice die in 1–3 days (type 14 may be avirulent for mice).
- **Antigen detection:** SSS in CSF by precipitation or **latex agglutination**; capsular polysaccharide in blood, urine, CSF by **CIE**; **immunochromatographic urine antigen test**.
- **Biomarkers:** **CRP** (latex agglutination) and **procalcitonin** (prognosis, response). **PCR** — useful when antibiotics already given (esp. meningitis).

11. PROPHYLAXIS & TREATMENT

- Immunity is **type-specific** (anticapsular antibody); ~90 serotypes make a complete vaccine impracticable.

Vaccine	Details
23-valent polysaccharide vaccine	80–90% protection; for high-risk persons — asplenia, sickle cell disease, celiac disease, chronic renal/lung/heart/liver disease, diabetes, HIV. Not for children < 2 years
7-valent conjugate vaccine	Conjugated to CRM197 protein of <i>C. diphtheriae</i> ; for children 2 months to 2 years

- Treatment:** parenteral **penicillin** (severe), **amoxicillin** (mild) if sensitive. Resistance is due to **altered penicillin-binding proteins, not β -lactamase**; resistant strains often multidrug-resistant (**DRSP**, originated in Spain).
- Resistant strains: **third-generation cephalosporin**; **vancomycin** reserved for life-threatening infections with highly resistant strains.

CLINICAL CORRELATION

- Elderly patient, rigors, rusty sputum, lobar consolidation → **pneumococcal lobar pneumonia**.
- Child with fever, neck rigidity, CSF with Gram-positive diplococci in polymorphs → **pneumococcal meningitis**.
- Splenectomised / sickle cell patients are highly susceptible → give pneumococcal vaccine.

MEMORY AID

- Pneumococcus identification — "**BOIL**": **B**ile soluble, **O**ptochin sensitive, **I**nulin fermented, **L**anceolate diplococci (+ Quellung).
- Pneumococcal penicillin resistance = **PBP change**, not β -lactamase (same principle as MRSA).

GOLDEN ONE-LINERS

- Draughtsman / carrom coin colonies — **autolysis** of *S. pneumoniae*.
- Bile solubility due to **autolytic amidase** (alanine–muramic acid bond).
- Optochin = **ethyl hydrocupreine**.
- Quellung reaction — **Neufeld (1902)**; transformation — **Griffith (1928)**.
- CRP is an **acute-phase protein** from hepatocytes, not an antibody.
- Most virulent serotype — **type 3**; most serious infection — **meningitis**.
- Conjugate vaccine carrier protein — **CRM197**.

EXAM FOCUS

- Differences between *S. pneumoniae* and viridans streptococci (Table 23.1) — very frequently asked.
- Bile solubility and optochin tests — principle and procedure.
- Laboratory diagnosis of pneumococcal pneumonia and meningitis.
- Quellung reaction and vaccines.

EXAM QUESTIONS (from the textbook)

Essays

- Describe the culture and identification of *Streptococcus pneumoniae*.
- Describe the laboratory diagnosis of pneumococcal pneumonia.
- Describe the antigenic properties of the capsule and methods of typing.
- Enumerate the organisms causing meningitis and write the laboratory diagnosis of meningitis.

Short answers / Short notes

- Quellung reaction; bile solubility test; optochin sensitivity
- Virulence factors of *S. pneumoniae*
- Biochemical tests for the identification of *S. pneumoniae*
- Vaccines against *S. pneumoniae*